# Actuation-Aware Sensor Validation for Fault-Resilient Closed-Loop Hydroponic Control

**Abstract**

Closed-loop hydroponic systems can convert transient sensor faults into unnecessary dosing, cooling, pumping, or carbon dioxide actuation. This paper presents a fault-resilient hydroponic control system whose central method is AASVR, an actuation-aware sensor validation and rectification layer. AASVR combines robust range and rate gates, uncommanded-trend checks, persistence states, bounded rectification, and actuation authorization. The system is evaluated on hydroponic deployment data with injected faults and on public HAI and SKAB process datasets. In hydroponic replay, AASVR achieved 0.859 balanced accuracy and reduced mean false actuation to 0.145 events per trial. The results support actuation-aware validation as a practical safety layer for low-cost hydroponic automation.

*Keywords:* sensor validation, data rectification, fault detection, process control, hydroponics, actuation

## 1. Introduction

Hydroponic controlled-environment systems depend on continuous measurement and actuation. In a recirculating Nutrient Film Technique (NFT) installation, the same data stream that describes pH, electrical conductivity (EC), air temperature, humidity, water temperature, water level, and carbon dioxide (CO2) can also trigger dosing pumps, chillers, valves, cooling devices, and circulation equipment. A faulty measurement is therefore not only a data-quality issue. It can become a physical command that perturbs the nutrient solution or growth environment.

This coupling is especially important in low-cost hydroponic deployments because sensors may spike, drift, saturate, drop out, or respond slowly after a real actuator command. A single high-pH spike should not immediately authorize acid dosing; a temporary EC excursion should not be treated the same as a persistent concentration shift; and water-level values without defensible calibration should not be allowed to dominate the control analysis.

These examples motivate a validation layer that is aware of the downstream actuation decision.

Conventional filtering methods are useful, but they usually ask whether a measurement looks noisy or anomalous in the signal domain. They do not directly ask whether the value is trustworthy enough to update the control state or authorize an actuator. We therefore introduce *actuation-aware sensor validation and rectification* as the central problem for fault-resilient hydroponic control. The key observation is that a value may be worth logging for diagnosis while still being too untrusted to drive a pump, valve, or cooler.

We present a closed-loop NFT hydroponic system as the primary engineered platform and propose AASVR as its central fault-resilience method. AASVR combines robust plausibility gates, persistence states, bounded rectification, and explicit actuation authorization. Public HAI and SKAB process datasets are used as external stress tests for the same validation interface, but they are not used to make agronomic claims.

The contributions are as follows. First, we describe an integrated closed-loop hydroponic sensing and control platform with separate growth and nutrient-solution modules. Second, we formulate sensor-fault handling in that platform as an actuation-aware validation and rectification problem. Third, we implement AASVR as a streaming state-machine method with bounded updates and explicit actuation authorization. Fourth, we evaluate the method on real hydroponic time series with synthetic faults, ablations, and public external benchmarks. Fifth, we provide a reproducible Docker and LaTeX workflow with dataset descriptors and paper-generation artifacts.

## 2. Related Work

### *2.1. Sensor faults in closed-loop systems*

Sensor data-quality reviews distinguish missingness, outliers, drift, stuck values, saturation, and inconsistent rates as recurring faults in deployed systems [1]. Agricultural IoT systems face the same problem when sensor failure or miscalibration propagates into monitoring or control decisions [2, 3]. In closed-loop hydroponics, these faults have downstream consequences because an erroneous measurement can change the actuator command. This motivates treating validation, rectification, and actuation as separate decisions rather than as one smoothing step.

### 2.2. Filtering and monitoring baselines

The baseline family used in this paper covers common online filters, process-monitoring methods, and lightweight one-class anomaly detectors. Hampel-style robust outlier detection uses robust scale estimates to suppress isolated deviations [4]. Kalman filtering gives a probabilistic recursive estimator for linear dynamical systems [5]. EWMA and CUSUM-style monitoring are standard approaches for persistent shifts [6, 7]. Multivariate PCA process monitoring uses Hotelling's $T^2$ and residual statistics to capture coordinated changes across variables [8, 9]. Isolation Forest, One-Class SVM, and local outlier factor add common one-class machine-learning comparisons for anomaly detection [10, 11, 12]. AASVR is complementary to these methods because it adds a control-facing authorization decision.

### 2.3. External process datasets

Public process datasets provide independent stress tests for measurement validation. HAI 21.03 contains industrial-control attack data from a hardware-in-the-loop testbed [13]. SKAB contains multivariate process-loop experiments with anomaly and changepoint labels [14]. SWaT, WaDi, and Tennessee Eastman are relevant larger benchmarks, but the current paper reports only the public benchmark paths that are already processed in this repository [15, 16, 17].

## 3. Hydroponic System Design and Deployment

The target platform is a closed NFT hydroponic system for lettuce growth. NFT systems recirculate a shallow nutrient stream through sloped growth channels, allowing water and nutrient reuse when solution management is reliable [18, 19]. The deployed system is organized into two physical modules. The Growth Module houses the plants, lighting, air-temperature and humidity sensing, CO2 sensing, visual monitoring, and environmental actuation. The Nutrient Solution Container Module stores and recirculates the nutrient solution and contains the water-quality sensors and nutrient-solution actuators.

The Growth Module uses a DHT22 sensor for air temperature and humidity and an MH-Z19C nondispersive infrared CO2 sensor for gas concentration. CO2 regulation is implemented through a normally closed pneumatic solenoid valve connected to a CO2 cylinder. Air-temperature regulation is

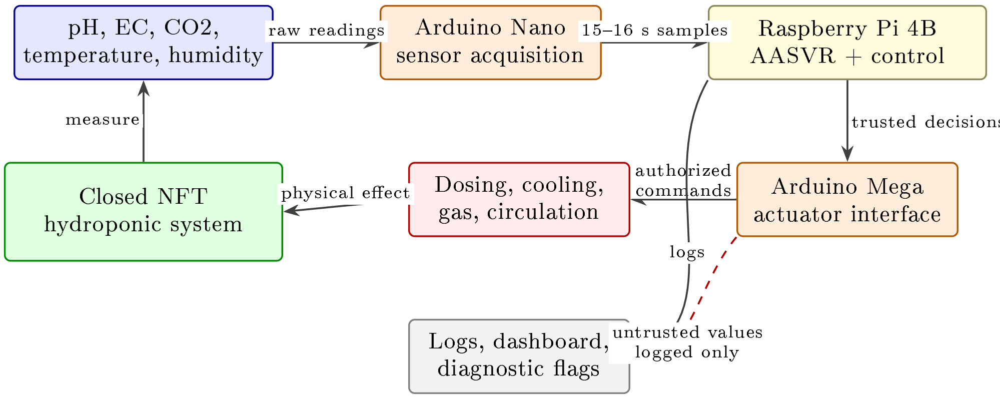


Figure 1: Hydroponic system architecture used as the primary deployment platform. Sensors in the Growth Module and Nutrient Solution Container Module are acquired through the Arduino Nano and sent to the Raspberry Pi 4B at approximately 15-second cadence. The Raspberry Pi logs data, updates the dashboard, runs the validation/control logic, and sends only authorized commands to the Arduino Mega actuator interface. AASVR is positioned between raw sensing and actuation so that untrusted values can be logged without directly triggering pumps, valves, chillers, or cooling devices.

supported by a thermoelectric cooler. Two 40 W LED grow-light arrays provide a controlled light cycle, and a camera provides visual monitoring of the crop during the growth cycle.

The Nutrient Solution Container Module uses an Atlas Scientific analog pH kit, an Atlas Scientific Mini Conductivity K 1.0 kit for EC, a water-temperature sensor, and a JSN-SR04T waterproof ultrasonic distance sensor for water level. pH correction is implemented through acid/base dosing, and EC correction through concentrated nutrient solution or distilled-water addition. After a pH or EC dosing action, the dosing subsystem is disabled for a 10-minute mixing interval before further dosing is allowed. A refrigerative water chiller regulates nutrient-solution temperature, an aerator runs throughout the growth cycle to support dissolved oxygen, and a water pump recirculates nutrient solution through the NFT channels.

The control unit connects sensors to an Arduino Nano and actuators to an Arduino Mega. Both microcontrollers communicate with a Raspberry Pi 4 Model B, which serves as the central control computer. This sensor-microcontroller-edge-computer structure is consistent with prior hydroponic automation systems, but the present paper focuses on the measurement-to-actuation reliability layer rather than on dashboarding as a contribution

Figure 2: Physical hydroponic system during operation. The photograph is included to show that the evaluation data come from a deployed closed-loop apparatus rather than from a purely simulated process.

[20, 21]. The Raspberry Pi receives median-filtered sensor readings from the Arduino Nano, logs the readings, updates the ThingSpeak-backed dashboard, applies the validation/control logic, and sends authorized actuation commands to the Arduino Mega. The dashboard and camera feed are retained as deployment infrastructure; they are not claimed as the research contribution of this paper.

Table 1: Hydroponic platform components relevant to measurement validation and actuation. Water level is listed because it is part of the deployed system, but it is excluded from the primary quantitative analysis because the available raw values are not defensibly calibrated.

| Subsystem | Measurements | Actuation or use |
|---|---|---|
| Growth module | Air temperature, humidity | Thermoelectric cooling |
| Growth module | CO2 concentration | CO2 solenoid valve |
| Growth module | Camera feed | Visual crop monitoring |
| Growth module | Light schedule | LED grow lights |
| Solution module | pH | Acid/base dosing, 10 min lockout |
| Solution module | EC | Nutrient/water dosing, 10 min lockout |
| Solution module | Water temperature | Refrigerative chiller |
| Solution module | Water level | Warning only in this analysis |
| Solution module | Recirculation state | Pump and aerator operation |

This architecture makes sensor validation a control-safety problem. A conventional smoother can reduce noise, but it does not decide whether a measurement is safe enough to authorize acid dosing, nutrient dosing, CO2 release, or cooling. In the deployed hydroponic system, a false high-pH event can trigger acid dosing, a false EC event can trigger nutrient or dilution dosing, a false water-temperature event can trigger chiller operation, and a false CO2 event can open the gas valve. The problem is therefore not only detecting anomalous samples; it is preventing untrusted samples from becoming physical corrections. AASVR is introduced as a layer between the sensing loop and the actuator loop, rather than as an offline data-cleaning step.

## 4. Problem Formulation

For hydroponic sensor $s$ at time $t$, let $y_s(t)$ be the raw measurement and $\hat{x}_s(t)$ be the trusted value exposed to control. The method observes recent history $Y_s(t - w : t)$, broad physical limits $[P_{s,\min}, P_{s,\max}]$, a control band

$[L_s^a, U_s^a]$, a rate limit $r_s$, and any available actuator state $a(t)$. In this system, $s$ may correspond to pH, EC, humidity, air temperature, water temperature, or CO2, while $a(t)$ may represent dosing, cooling, circulation, or gas-release activity.

Table 2: Notation used in the AASVR formulation. Variables are defined for one sensor stream and then applied independently or jointly across the hydroponic sensor set.

| Symbol | Meaning |
|---|---|
| $y_s(t)$ | Raw value from sensor $s$ at sample $t$ |
| $\hat{x}_s(t)$ | Trusted value exposed to the controller |
| $Y_s(t-w:t)$ | Rolling window of recent raw values |
| $[P_{s,\min}, P_{s,\max}]$ | Broad physically plausible range |
| $[L_s^a, U_s^a]$ | Actuation or control band |
| $r_s$ | Maximum plausible rate of change |
| $q_s(t)$ | Trust score in $[0,1]$ |
| $z_s(t)$ | Sensor state in the AASVR state set |
| $A_s(t)$ | Binary actuation authorization flag |
| $\theta$ | Tunable AASVR parameter vector |

We define a measurement as *untrusted* when it is missing, physically implausible, rate-inconsistent, inconsistent with recent trusted state, or inconsistent with expected actuator effects. We define a *false actuation* as an authorization caused by untrusted evidence. A *missed actuation* is a failure to authorize action when the trusted state remains outside the control band.

The validation decision is separated from the control decision. Let $g_s(t) \in \{0,1\}$ denote whether a raw sample passes all measurement gates, and let $z_s(t)$ denote one of five states: normal, suspect transient, persistent deviation, accepted regime shift, and fault alert. The controller receives $\hat{x}_s(t)$ and $A_s(t)$, not the raw value alone. This separation is necessary because a value can be physically possible but still too uncertain to authorize actuation.

The evaluation follows a control-aware objective,

$$\begin{aligned} J(\theta) = &w_1 E_{\text{pass}} + w_2 E_{\text{reject}} + w_3 A_{\text{false}} + w_4 A_{\text{missed}} \\ &+ w_5 D_{\text{delay}} + w_6 T_{\text{unsafe}} + w_7 C_{\text{compute}}, \end{aligned} \tag{1}$$

where the terms respectively measure anomalous pass-through, false rejection, false actuation, missed actuation, detection delay, residence outside safe bands, and computational cost.

## 5. Proposed Method: AASVR

### *5.1. Overview*

Figure 3 summarizes the AASVR pipeline. The method separates four decisions that are often conflated: measurement validation, trusted-state update, rectification, and actuation authorization. This separation is the core design choice. In the hydroponic system, this means that a suspicious pH value can be stored with a diagnostic flag while acid/base dosing remains suppressed until the value is persistent and trusted.

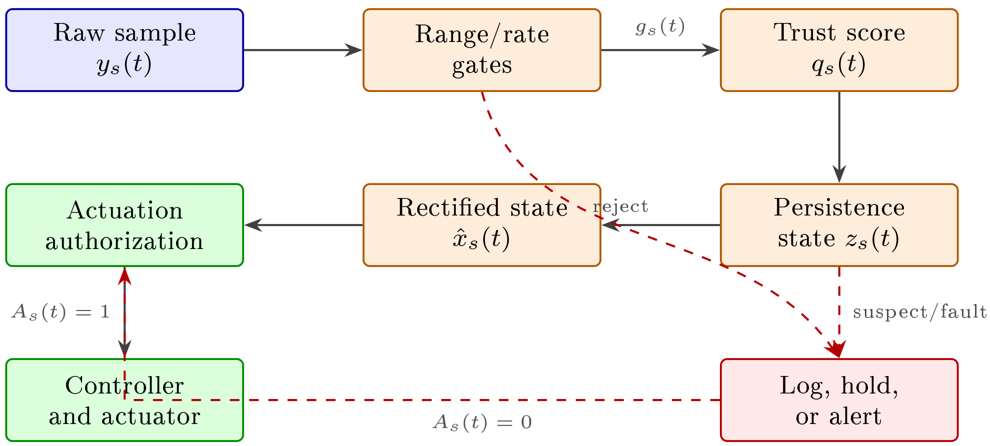


Figure 3: Architecture of AASVR. Raw measurements first pass through robust plausibility gates that check physical range, local rate, trusted-state deviation, and uncommanded trends. Accepted or suspect readings then update a persistence state machine, which decides whether the trusted state should be updated, held, or rectified by a bounded prediction. Only trusted out-of-band states can authorize actuation. Rejected or fault-alert readings are logged and may trigger maintenance alerts, but they do not directly drive the actuator.

### *5.2. Robust plausibility gate*

For each sensor, AASVR estimates a robust scale over adjacent changes,

$$\sigma_{\Delta,s}(t) = 1.4826\,\mathrm{MAD}\left(\Delta y_s(t-w:t)\right). \tag{2}$$

The accepted deviation tolerance is

$$\xi_s(t) = \max(\xi_{s,\min}, c_s\sigma_{\Delta,s}(t), r_s\Delta t, u_s). \tag{3}$$

A reading must satisfy broad physical range, maximum rate, and trusted-state deviation gates. The three elementary gates are

$$\begin{aligned} G_s^P(t) &= \mathbf{1}\{P_{s,\min} \le y_s(t) \le P_{s,\max}\}, \\ G_s^R(t) &= \mathbf{1}\{|y_s(t) - y_s(t-1)|/\Delta t \le r_s\}, \\ G_s^D(t) &= \mathbf{1}\{|y_s(t) - \hat{x}_s(t-1)| \le \xi_s(t)\}. \end{aligned} \tag{4}$$

A separate trend guard $G_s^T(t)$ rejects slow monotonic movement when no relevant actuator activity explains the direction of change. The overall gate is therefore

$$g_s(t) = G_s^P(t) G_s^R(t) G_s^D(t) G_s^T(t) G_s^M(t), \tag{5}$$

where $G_s^M(t)$ is one only when the value is present and parseable. The broad physical range is deliberately different from the crop-control band: for example, a pH value can be undesirable for lettuce while still being physically possible and therefore not automatically a sensor fault.

### *5.3. Trust score*

AASVR also computes a continuous trust score so that control can distinguish weakly plausible values from strongly confirmed values. The implementation uses a weighted additive score,

$$q_s(t) = \sum_{j \in \mathcal{Q}} w_j q_{j,s}(t), \quad \sum_{j \in \mathcal{Q}} w_j = 1, \tag{6}$$

where $\mathcal{Q}$ contains range, rate, persistence, actuator-consistency, missingness, and trend components. Each component is scaled to $[0, 1]$. In the hydroponic controller, $q_s(t)$ is used together with persistence and cooldown logic rather than as a standalone anomaly score.

### *5.4. State-machine persistence logic*

Figure 4 shows the five persistence states. A single failed gate moves a sensor into a suspect transient state. Repeated failures create a persistent-deviation state. Consistent evidence can be accepted as a new process regime, while missing, stuck, saturated, or contradictory readings escalate to fault alert. This matters for hydroponics because real mixing and thermal responses can take multiple samples after dosing or cooling, whereas isolated spikes should not immediately change the trusted state.

The state update can be written as

$$z_s(t) = \mathcal{T}\left(z_s(t-1), g_s(t), q_s(t), h_s(t), a(t)\right), \tag{7}$$

where $h_s(t)$ summarizes persistence-window evidence. This representation makes the state machine explicit without treating every transition as a separate hand-tuned rule in the manuscript. The transition function $\mathcal{T}$ is implemented by Algorithm 1.

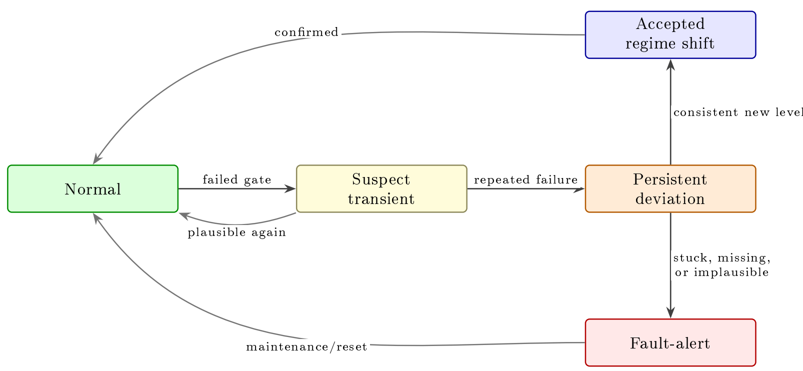


Figure 4: AASVR state machine. The state machine prevents one failed sample from immediately becoming a maintenance alert, but also prevents repeated failed samples from silently entering the controller. The accepted-regime-shift path allows real process changes to be incorporated after confirmation. The fault-alert path blocks single-sensor actuation authorization until maintenance, reset, or independent evidence restores trust.

### *5.5. Rectification and actuation authorization*

When a reading is accepted, AASVR sets $\hat{x}_s(t) = y_s(t)$. When it is suspect, the method either holds $\hat{x}_s(t-1)$ or applies a bounded update limited by $r_s \Delta t$:

$$\hat{x}_s(t) = \begin{cases} y_s(t), & z_s(t) \in \{\text{normal}, \text{accepted}\}, \\ \hat{x}_s(t-1), & z_s(t) = \text{suspect}, \\ \hat{x}_s(t-1) + \text{clip}(\Delta y_s, -r_s\Delta t, r_s\Delta t), & \text{bounded}, \\ \text{alert}, & z_s(t) = \text{fault}. \end{cases} \tag{8}$$

Actuation is authorized only if the trusted value is outside the control band, the violation persists for the configured number of samples, the sensor state is

**Algorithm 1** Streaming AASVR update for one sensor

**Require:** Raw value $y_s(t)$, previous trusted state $\hat{x}_s(t-1)$, recent history, limits, actuator state.

Estimate robust local scale $\sigma_{\Delta,s}(t)$ and tolerance $\xi_s(t)$.
Apply physical-range, rate, trusted-deviation, missingness, and trend gates.
Update the persistence state from the gate outcome.
**if** the reading is accepted **then**
    Set $\hat{x}_s(t) \leftarrow y_s(t)$.
**else if** the reading is transient suspect **then**
    Hold or bounded-predict $\hat{x}_s(t)$.
**else**
    Keep the last safe estimate and log a fault-alert decision.
**end if**
Authorize actuation only if trusted state, persistence, score, and cooldown rules are satisfied.
**return** Trusted value, sensor state, trust score, actuation flag, and log record.

normal or accepted-regime-shift, the trust score is sufficient, and the actuator is not in cooldown:

$$\begin{aligned} A_s(t) = &\mathbf{1}\{\hat{x}_s(t) \notin [L_s^a, U_s^a]\}\mathbf{1}\{p_s(t) \geq k_s\}\mathbf{1}\{q_s(t) \geq q_{\min}\} \\ &\times \mathbf{1}\{z_s(t) \in \mathcal{Z}_{\text{safe}}\}\mathbf{1}\{\ell_s(t) = 0\}. \end{aligned} \tag{9}$$

Here $p_s(t)$ is the number of consecutive trusted control-band violations, $k_s$ is the persistence requirement, $\mathcal{Z}_{\text{safe}} = \{\text{normal}, \text{accepted}\}$, and $\ell_s(t)$ is the actuator lockout indicator. For pH and EC, this rule complements the 10-minute dosing lockout already present in the deployed system; for CO2 and temperature, it prevents transient readings from becoming immediate gas-release or cooling commands.

## 6. Evaluation Protocol

### *6.1. Hydroponic deployment data*

The hydroponic data contain 121244 raw rows collected from 2024-02-27 to 2024-03-28 at a median cadence of approximately 16 seconds. The primary analysis uses EC, pH, humidity, air temperature, water temperature, and CO2. Water level is part of the deployed system but is excluded from the primary analysis because the available raw values are not defensibly calibrated.

### *6.2. External benchmarks*

HAI 21.03 and SKAB are used as public external stress tests. HAI labels are timestamp-level attack indicators over an industrial-control testbed. SKAB labels are timestamp-level anomaly indicators over process-loop experiments. For these datasets, a timestamp is predicted abnormal when at least 30% of sensor streams reject or alert. We also report a threshold sensitivity sweep from 10% to 90%.

### *6.3. Baselines and metrics*

The baselines are raw thresholding, the original manuscript method, moving average, moving median, Hampel, Kalman, EWMA, CUSUM, PCA-style monitoring, Isolation Forest, One-Class SVM, and local outlier factor. The one-class machine-learning baselines are calibrated on the prefix of each replay window and then evaluated on the same injected-fault segment as AASVR. The hydroponic synthetic-fault benchmark reports precision, recall, specificity, balanced accuracy, event recall, detection delay, false alarm events, and false actuation. Native-label external benchmarks report timestamp-level precision, recall, specificity, and balanced accuracy.

Table 3: Baseline taxonomy used in the hydroponic benchmark. The final column marks whether a method explicitly separates anomaly detection from actuation authorization.

| Family | Methods | Primary role | Act.-aware |
|---|---|---|---|
| Rule | Raw threshold, original method | Existing hydroponic control logic | No |
| Filters | Moving average, moving median, Hampel, Kalman | Noise and spike suppression | No |
| Process monitoring | EWMA, CUSUM, PCA-style residual monitoring | Shift or process-deviation detection | No |
| One-class ML | Isolation Forest, One-Class SVM, LOF | Data-driven anomaly boundary | No |
| Proposed method | AASVR | Validation, rectification, and authorization | Yes |

## 7. Results

### *7.1. Hydroponic deployment characterization*

The deployment data are first used to characterize the real measurement background on which AASVR operates. The 121244-row feed covers a continuous 30-day lettuce growth period, with approximately 16-second median sampling. The key point for this paper is not that the dashboard collected data, but that the closed-loop system generated the kind of sensor-to-actuator stream in which a bad reading can trigger a physical correction.

### *7.2. Hydroponic synthetic-fault benchmark*

Table 5 summarizes the primary target-domain result. AASVR achieved the highest balanced accuracy and the lowest mean false-actuation count among the compared methods. LOF reached similar balanced accuracy by increasing recall, but it produced far more false actuation. One-Class SVM had the highest recall but poor specificity. This is the intended comparison: AASVR is not claimed to maximize every signal-level metric, but to improve the control-facing tradeoff between rejection and unnecessary actuation.

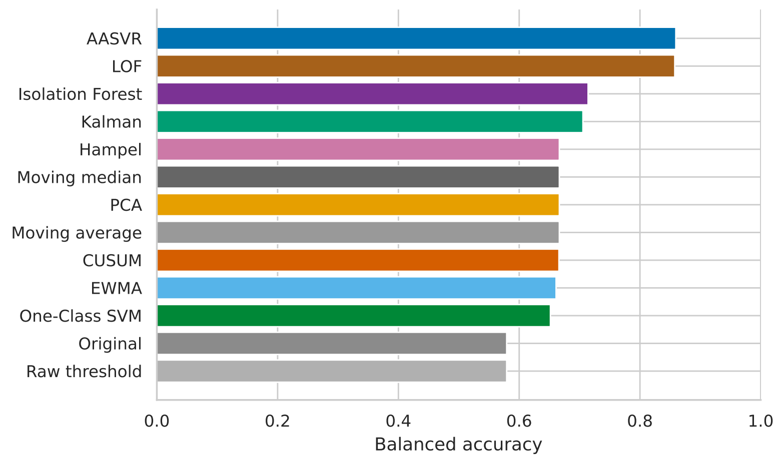


Figure 5: Hydroponic synthetic-fault balanced accuracy across methods. The comparison is computed on repeated injected faults over windows sampled from the real hydroponic data. AASVR ranks first because it maintains high recall while preserving enough specificity to avoid passing many injected faults into the controller.

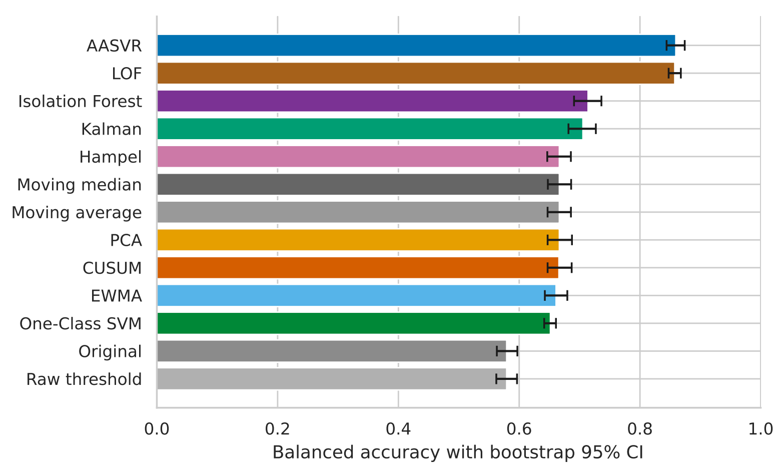


Figure 6: Bootstrap uncertainty for hydroponic synthetic-fault balanced accuracy. AASVR reached a mean balanced accuracy of 0.859 with a 95% bootstrap confidence interval of 0.844–0.874. The interval separates AASVR from the conventional baselines in the current benchmark, but the result remains tied to the injected fault model and available hydroponic trace.

Table 4: Hydroponic measurement streams used in the primary AASVR analysis. The table separates deployed sensing from quantitative use so that the water-level exclusion is explicit rather than hidden.

| Variable | Module | Primary analysis | Control relevance |
|---|---|---|---|
| EC | Solution | Yes | Nutrient/water dosing evidence |
| pH | Solution | Yes | Acid/base dosing evidence |
| Humidity | Growth | Yes | Growth-environment monitoring |
| Air temperature | Growth | Yes | Air-cooling evidence |
| Water temperature | Solution | Yes | Chiller evidence |
| CO2 | Growth | Yes | CO2 valve evidence |
| Water level | Solution | No | Warning channel; calibration unresolved |

Table 5: Hydroponic synthetic-fault benchmark. The table reports leading methods by balanced accuracy on injected faults over the real hydroponic signal background. Bold entries mark the best value in each metric column among the displayed methods; lower is better for false actuation.

| Method | Recall | Specificity | Bal. acc. | FPR | False act. |
|---|---|---|---|---|---|
| AASVR | 0.839 | 0.880 | **0.859** | 0.120 | **0.145** |
| LOF | 0.935 | 0.780 | 0.858 | 0.220 | 2.752 |
| Isolation Forest | 0.545 | 0.884 | 0.714 | 0.116 | 2.805 |
| Kalman | 0.591 | 0.820 | 0.706 | 0.180 | 3.029 |
| Hampel | 0.353 | 0.980 | 0.667 | 0.020 | 3.038 |
| CUSUM | 0.344 | 0.987 | 0.666 | 0.013 | 3.060 |
| One-Class SVM | **0.953** | 0.350 | 0.652 | 0.650 | 2.757 |
| Original | 0.161 | **0.998** | 0.579 | **0.002** | 3.060 |

### *7.3. Sensor-wise and fault-type comparisons*

Table 6 breaks down AASVR performance by hydroponic sensor. CO2, air temperature, and water temperature faults are detected with balanced accuracy above 0.88. EC is the weakest stream, mainly because gradual concentration faults are harder to distinguish from plausible nutrient-solution changes without an independent reference.

Table 7 and Figure 7 show performance by injected fault class. AASVR performs best on dropout, step, spike, bias, multi-spike, and noise-burst faults. Drift remains the hardest case because it can stay inside physical range and rate limits for long periods.

Table 6: AASVR synthetic-fault performance by hydroponic sensor. False actuation is reported as mean events per injected-fault trial.

| Sensor | Recall | Specificity | Bal. acc. | False act. |
|---|---|---|---|---|
| CO2 | **0.944** | 0.881 | **0.913** | **0.036** |
| Air temperature | 0.912 | 0.877 | 0.895 | 0.060 |
| Water temperature | 0.902 | 0.876 | 0.889 | 0.048 |
| pH | 0.840 | **0.885** | 0.862 | 0.060 |
| EC | 0.596 | 0.880 | 0.738 | 0.524 |

Table 7: AASVR synthetic-fault performance by injected fault type. Drift is the most difficult condition because slow monotonic movement can resemble a real process change.

| Fault type | Recall | Specificity | Bal. acc. | False act. |
|---|---|---|---|---|
| Dropout | **1.000** | **0.886** | **0.943** | **0.000** |
| Step | 0.890 | 0.881 | 0.886 | 0.167 |
| Spike | 0.883 | 0.878 | 0.880 | 0.200 |
| Bias | 0.876 | 0.877 | 0.876 | 0.200 |
| Multi-spike | 0.874 | 0.878 | 0.876 | **0.000** |
| Noise burst | 0.856 | 0.882 | 0.869 | 0.133 |
| Drift | 0.494 | 0.877 | 0.686 | 0.317 |

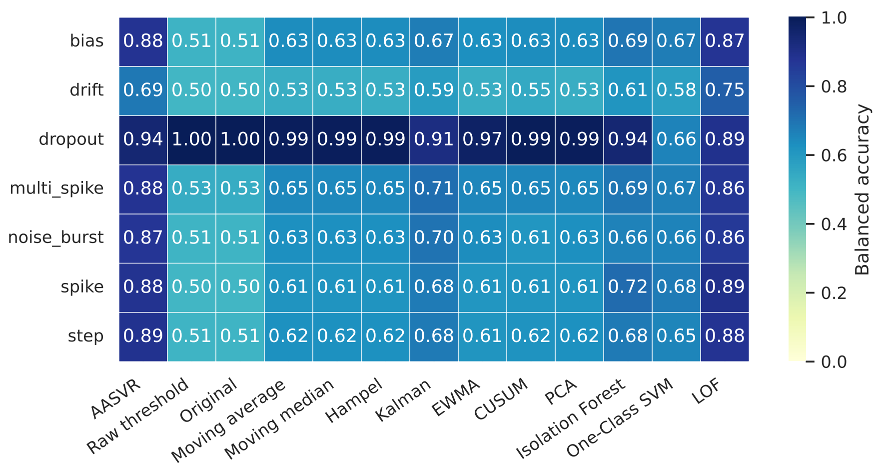


Figure 7: Hydroponic synthetic-fault heatmap by sensor and fault type. The figure summarizes where AASVR is strongest and where slow drift remains a limitation.

### *7.4. Ablation and control effect*

Table 8 shows that several ablations preserve the same detection score but change control behavior. Removing actuation confirmation increases false actuation from 0.145 to 0.569 events per trial. This is the clearest evidence that AASVR is not only a detector: the actuation-aware layer changes downstream control outcomes.

The parameter sweep in Table 9 shows that the selected configuration is not an isolated optimum. Multiple trust-score and transient-limit settings produced the same headline objective when the robust scale multiplier was 5.0, while reducing the scale multiplier to 4.0 slightly lowered the control-aware objective.

### *7.5. Representative hydroponic trace*

Figures 8 and 9 show replay segments from the hydroponic deployment. The traces are included to connect the aggregate benchmark to the actual deployment setting. They illustrate the measurement-to-state transformation that precedes any actuation decision.

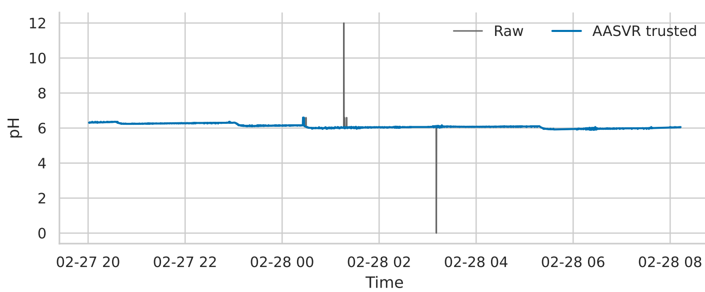


Figure 8: Representative pH replay from the hydroponic deployment data. The raw stream and AASVR trusted estimate are plotted over an early segment of the experiment. The figure is not used as a ground-truth fault label; it is included to show how the method exposes a trusted state rather than passing every raw measurement directly to the controller.

Table 8: AASVR ablation summary. Detection scores can remain similar while control-safety outcomes change. In particular, removing actuation confirmation leaves balanced accuracy unchanged but increases false actuation, which supports evaluating control-aware metrics rather than point-wise detection alone.

| Variant | Bal. acc. | Recall | False act. |
|---|---|---|---|
| Full AASVR | **0.859** | 0.839 | 0.145 |
| No actuation confirmation | **0.859** | 0.839 | 0.569 |
| No cooldown | **0.859** | 0.839 | 0.269 |
| No robust MAD scale | 0.854 | **0.864** | **0.133** |
| No trend guard | 0.843 | 0.801 | 0.260 |
| Bounded prediction mode | 0.810 | 0.734 | 0.476 |

Table 9: Top AASVR parameter settings from the hydroponic synthetic-fault sweep. The selected setting uses $c_s = 5.0$, $q_{\min} = 0.70$, and transient limit 2.

| $c_s$ | $q_{\min}$ | Transient limit | Bal. acc. | Objective |
|---|---|---|---|---|
| 5.0 | 0.70 | 2 | **0.859** | **0.783** |
| 5.0 | 0.70 | 1 | **0.859** | **0.783** |
| 5.0 | 0.50 | 3 | **0.859** | **0.783** |
| 5.0 | 0.85 | 2 | **0.859** | **0.783** |
| 4.0 | 0.70 | 3 | 0.859 | 0.782 |

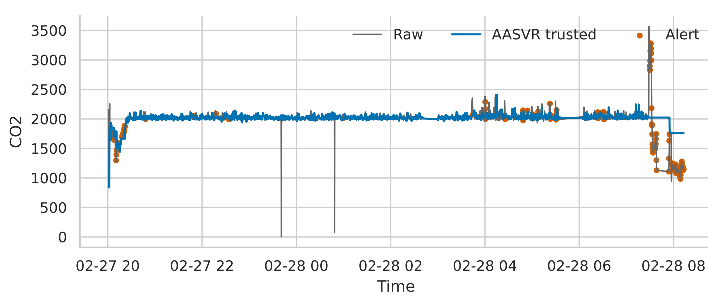


Figure 9: Representative CO2 replay from the hydroponic deployment data. The trace shows the same trusted-state interface on an environmental-control stream, where transient raw readings should not directly authorize gas-release commands.

### *7.6. External benchmark stress tests*

Table 10 reports the fixed 30% native-label protocol for HAI and SKAB. These datasets are included after the hydroponic results because they are supporting stress tests, not replacements for the target system. AASVR ranks first on both datasets under this fixed rule, but the nature of the result differs. HAI has high specificity and modest recall, while SKAB has high recall and low specificity.

Table 10: External native-label results under the fixed 30% timestamp aggregation rule. The values should be read as cross-domain stress-test evidence rather than direct hydroponic fault evidence. HAI contains cyber-physical attack labels, while SKAB contains process-loop anomaly labels.

| Dataset | Method | Precision | Recall | Specificity | Bal. acc. |
|---|---|---|---|---|---|
| HAI | AASVR | 0.078 | **0.404** | 0.929 | **0.667** |
| HAI | Kalman | **0.136** | 0.312 | **0.971** | 0.641 |
| SKAB | AASVR | **0.379** | **0.975** | **0.358** | **0.666** |
| SKAB | Kalman | 0.376 | **0.975** | 0.349 | 0.662 |

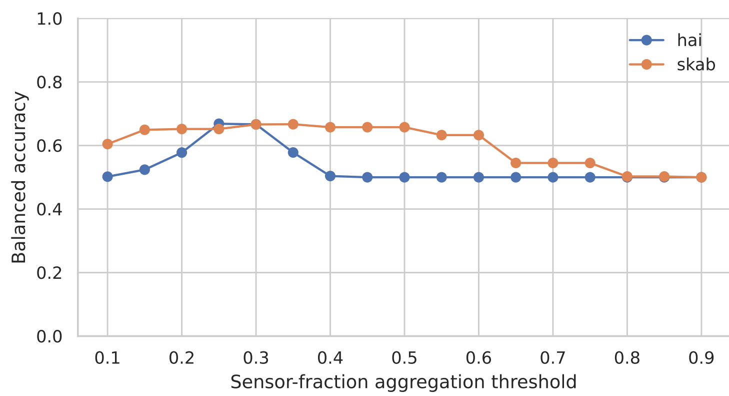


Figure 10: Threshold sensitivity for timestamp-level external labels. The fixed 30% aggregation rule is the headline protocol, but this plot shows how AASVR changes when the required fraction of rejecting streams is varied. SKAB is stable around the headline setting, while HAI reveals a tradeoff between recall and specificity.

### *7.7. Agronomic operation evidence*

The lettuce growth trials are retained as deployment evidence, not as causal proof that AASVR improved crop yield. The automated treatment

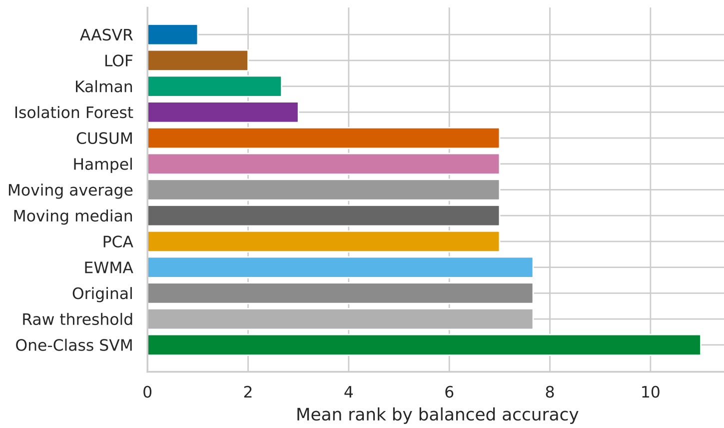


Figure 11: Mean rank by balanced accuracy across available benchmark tables. The ranking combines hydroponic synthetic faults with native-label external benchmark summaries. It is useful for orientation, but it should not be interpreted as a leaderboard because the datasets use different label semantics and evaluation units.

differed from the soil and manual DWC controls in enclosure, lighting, cooling, nutrient delivery, CO2 regulation, and automation. The agronomic comparison therefore shows that the system operated through complete growth cycles while supporting the method evaluation; it is not used to isolate the effect of the validation algorithm.

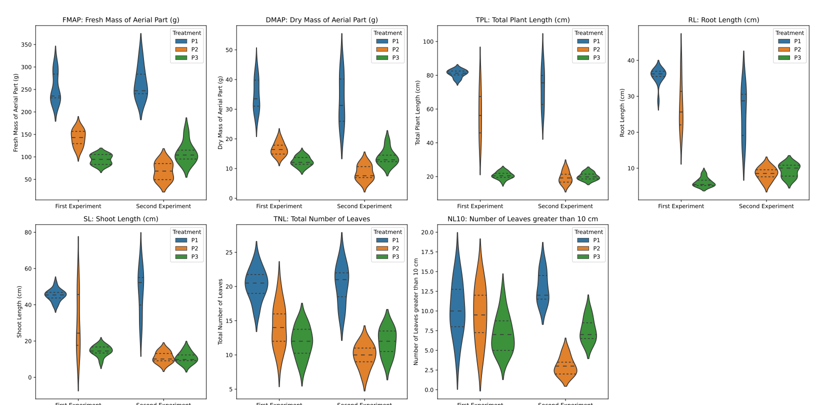


Figure 12: Agronomic measurements from the hydroponic deployment and comparison treatments. These data are included as secondary operation evidence for the closed-loop hydroponic platform. They are not interpreted as causal evidence that AASVR alone improved plant growth because multiple system-level factors differ between treatments.

## 8. Discussion

### 8.1. Why actuation awareness matters

The ablation results show why actuation-aware validation should be evaluated with control metrics. A detector can produce the same point-wise score while changing the number of authorized actuator events. For hydroponic systems, this distinction is practical because unnecessary acid/base dosing, nutrient dosing, CO2 release, or cooling can disturb the process even if the signal-level anomaly score appears acceptable.

### 8.2. Cross-domain generalization

The external benchmark results support a cautious generalization claim. AASVR can run on independent industrial-control and process-loop datasets with the same streaming interface. However, HAI attacks and SKAB process anomalies are not identical to low-cost hydroponic sensor faults. The external results therefore test robustness of the validation principle, not agronomic performance.

### 8.3. Limitations

The hydroponic experiment does not provide independent reference instruments for every sensor at every time point. Synthetic fault injection is therefore necessary for controlled scoring, but injected faults cannot cover every real failure mode. The external datasets use timestamp-level labels, so the multivariate aggregation threshold affects the result. Water level is excluded from primary analysis because the available raw values are not defensibly calibrated. Crop outcomes are treated only as deployment context because enclosure, lighting, cooling, nutrient delivery, carbon dioxide regulation, and automation are confounded.

## 9. Conclusion

We presented a closed-loop hydroponic control system and proposed AASVR as its actuation-aware sensor validation and rectification layer. The method separates validation, state update, rectification, and actuation authorization so that untrusted measurements can be logged without directly driving physical commands. On hydroponic synthetic faults, AASVR improved balanced accuracy and reduced false actuation compared with conventional baselines. Public SKAB and HAI benchmarks show that the same interface transfers to external process data while exposing threshold-dependent tradeoffs that should be reported rather than hidden.

## Data Availability

The repository contains code, configuration files, processed-data descriptors, Docker workflow files, and paper-generation assets. Raw hydroponic and external benchmark files are not committed to the repository. Public datasets can be obtained from their original providers, and request-access datasets should be obtained under the providers' terms.

## Declaration of Competing Interest

The authors declare no competing interests for the current manuscript draft.

## Declaration of Generative AI and AI-Assisted Technologies

AI-assisted tools were used to help organize code, draft text, and check consistency. The authors remain responsible for the scientific content, claims, results, and final manuscript.

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